

Newton’ s cell action as a measure on cuts

After refereeing (GPT-6 Astra, 2026-09-28). A6: ACCEPT Proposition 6’ s Cameron–Martin norm, optimal equal-prior error and exact constant in the Planck mesh. RE-FINE the statement to two prescribed Gaussian bridge means with the same pinned endpoints and covariance, distinguish the variance rate from the diffusion coefficient, and identify the likelihood pairing as a Gaussian stochastic functional. The agreement with the mark model is structural: both have the sharp squared signal distance $2K_\tau/\hbar$ from the same Dirichlet energy. The bridge test attains its bound; the mark model approaches it by dense protocols. Both supply $\hbar > 0$, leaving necessity unchanged.

After refereeing (Fable, 2026-09-28). Theorems 1–2, Corollary 3’ s arithmetic and the quotations were re-derived and accepted. Corrections applied: Proposition 4 now names the Kullback–Leibler divergence, the shift δx , the phase K_τ/ε and the Cameron–Martin normalizer, and its link to Rivero 1998 is stated as a counterpart; Corollary 5 separates a floor on a cell’ s action from a floor on a cut’ s share (a factor $\frac{3}{4}$ for halving); § 3 labels the two-quarter-cut schedule a construction and adds the alternating reading favoured by 斲必半; the Lévy–Ciesielski comparison names the right quantity.

Result, 2026-09-27 (atlas § 1b, user remarks on arbitrary cuts and on the rod). Let a constant force F act on a body of mass m over a cell of duration τ , and let δx be the chord minus the parabola (the chord is the inertial comparison path with the same endpoints). Its Galileo action

$$K_\tau = \frac{m}{2} \int_0^\tau \dot{\delta x}^2 dt = \frac{F^2 \tau^3}{24m}$$

(the two-path note writes the same quantity as $(F/4v)A = \tau \Delta E/12$) behaves as a measure on the process of cutting the cell.

- **Theorem 1 (one cut).** A cut at fraction $s \in (0, 1)$ spends exactly $3s(1-s)K_\tau$, and this share equals the Cameron–Martin energy $\frac{m}{2} \int \dot{\phi}^2$ of the Schauder hat ϕ that the cut inserts, of height the sagitta $Fs(1-s)\tau^2/(2m)$.
- **Theorem 2 (any cut sequence).** The hats of successive cuts are orthogonal in the energy $\int \dot{f} \dot{g}$, so the spent shares add. Along any sequence of cuts, at any positions and in any order, the total spent equals $K_\tau - \sum_i K_{\tau_i}$ over the current pieces, and it tends to K_τ iff the mesh $\max_i \tau_i$ tends to zero; the remainder is at most $K_\tau \max_i \tau_i^2/\tau^2$.
- **Corollary 3 (the rod and the Mohist schedules).** Halving only the remaining piece each day, the schedule of the stick in Zhuangzi 33, spends $\frac{6}{7}K_\tau$, and so does one halving a day taken alternately from the front and the back. Removing a quarter from each end each day (two quarter-cuts a day, a construction motivated by the Mohist “taking from front and back”) spends $\frac{27}{28}K_\tau$. All three leave a definite share unspent forever, because they keep cut-off pieces of fixed length.
- **Proposition 4 (weight of a cut).** Under the Euclidean path measure of a free particle with variance rate $\hbar/m$ (Wiener measure), the force signal in one cut has Kullback–Leibler divergence exactly (spent share)/ $\hbar$. In real time the same share, divided by the action resolution, is the phase of the two-path comparison.
- **Proposition 6 (the mark mesh as a Cameron–Martin threshold; 2026-09-28, refereed ACCEPT with the hypotheses of § 5b).** Under the same path measure, the optimal equal-prior test between force F and no force in a cell, observing the whole path with pinned ends, errs with probability $\Phi(-\sqrt{K_\tau/2\hbar})$. It reaches ϵ iff $K_\tau \geq 2z_{1-\epsilon}^2 \hbar$, which is exactly the Planck paper’ s mark mesh $\tau_* = (48z_{1-\epsilon}^2 m \hbar / F^2)^{1/3}$, obtained there from real-time Gaussian marks with Robertson’ s bound.

- **Corollary 5 (a floor counts the cuts).** If every exhibited cut must spend at least an action $\kappa > 0$, at most K_τ/κ cuts can be exhibited, whatever the schedule. For halving refinement, level n is exhibited iff $\frac{3}{4}K_\tau 8^{-n} \geq \kappa$. The Planck paper's mark mesh $\tau_* = (48z_{1-\epsilon}^2 m\hbar/F^2)^{1/3}$ is the cell whose action is $2z_{1-\epsilon}^2\hbar$; its own halving spends $\frac{3}{2}z_{1-\epsilon}^2\hbar$.

Theorems 1 and 2 are elementary, and their content lies in the reading they give: Newton's refinement is a process that spends a fixed, finite action, with the nested additivity of the Lévy–Ciesielski construction of the Brownian bridge. Proposition 4 makes the correspondence exact: the action is the Cameron–Martin energy of the force's shift, decomposed over the hats. Corollary 5 is a consistency statement. It says where a floor, if present, stops the cutting, and leaves the necessity question of the fifth-postulate note where it was.

1. One cut

Put $C = F^2/(24m)$, so $K_\tau = C\tau^3$. On $[0, \tau]$ the parabola with acceleration F/m and the chord through its endpoints differ by

$$\delta x(t) = \frac{F}{2m} t(\tau - t), \quad \dot{\delta x}(t) = \frac{F}{m} \left(\frac{\tau}{2} - t \right),$$

$$\text{so } \frac{m}{2} \int_0^\tau \dot{\delta x}^2 dt = \frac{m}{2} \cdot \frac{F^2}{m^2} \cdot \frac{\tau^3}{12} = C\tau^3.$$

Cut at $t_c = s\tau$. The polygon through the three points $0, s\tau, \tau$ differs from the chord by the hat ϕ of height $h = \delta x(s\tau) = Fs(1-s)\tau^2/(2m)$, linear on $[0, s\tau]$ and on $[s\tau, \tau]$. The polygon differs from the parabola, on each piece, by the chord-minus-parabola of that piece: $\delta x = \phi + \delta x_1 + \delta x_2$ with δx_j supported on piece j and vanishing at its ends.

Orthogonality. $\dot{\phi}$ is constant on each piece and $\int_{\text{piece}} \dot{\delta x}_j = 0$, so $\int \dot{\phi} \dot{\delta x}_j = 0$; the δx_j have disjoint supports. Hence

$$K_\tau = \frac{m}{2} \int \dot{\phi}^2 + K_{s\tau} + K_{(1-s)\tau}.$$

The hat's energy. $\frac{m}{2} \int \dot{\phi}^2 = \frac{m}{2} h^2 \left(\frac{1}{s\tau} + \frac{1}{(1-s)\tau} \right) = \frac{mh^2}{2s(1-s)\tau} = \frac{F^2 s(1-s)\tau^3}{8m} = 3s(1-s)C\tau^3$. The same number follows from $\tau^3 = (s\tau)^3 + ((1-s)\tau)^3 + 3s(1-s)\tau^3$, which is the algebraic form of Theorem 1. The share is largest for halving, $\frac{3}{4}K_\tau$. $\square$

2. Any cut sequence

Each later cut acts inside one current piece, and Theorem 1 applies to that piece with its own s . The new hat is supported in the piece and vanishes at its ends; every earlier hat is linear there. The argument of §1 gives orthogonality to all earlier hats and to the other pieces' remainders. After any finite set of cuts, with pieces τ_i ($\sum \tau_i = \tau$),

$$K_\tau = \sum_{\text{cuts}} \frac{m}{2} \int \dot{\phi}_{\text{cut}}^2 + \sum_i K_{\tau_i}, \quad \sum_i K_{\tau_i} = C \sum_i \tau_i^3 \leq C\tau \max_i \tau_i^2.$$

The remainder vanishes iff the mesh does ($\sum_i \tau_i^3 \geq (\max_i \tau_i)^3$). The positions and the order enter only through which pieces are cut, never through the total. $\square$

This is the Schauder (Lévy–Ciesielski) expansion of δx in the Cameron–Martin space $H_0^1[0, \tau]$, adapted to an arbitrary nested cut sequence; for dyadic cuts it is the textbook expansion (Ciesielski 1961; metadata). The reachability note has the same cubic law for the area $2F^2t^3/(3m)$ of the reachable lens of a cut. The bridge's own conditional

variance shows the same nested additivity with a quadratic law: for diffusion constant D , $\int_0^L \text{Var}(X_u | \text{ends}) du = DL^2/6$, and a cut at fraction s removes $Ds(1-s)L^2/3$ of it.

3. The rod and the two Mohist schedules

Zhuangzi 33 (dialecticians' list, text): 一尺之捶，日取其半，萬世不竭， “a one-foot stick, each day take half, in ten thousand generations it is not exhausted.” Mohist Canon B (B60 in Graham' s numbering; canon): 非半弗斲，則不動，說在端， “without halving there is no cutting and no moving; the explanation lies in the point (端)” , with the explanation (text) 前則中無爲半，猶端也。前後取，則端中也 (working gloss: “going forward, at the middle nothing serves as a half: it is like a point; taking from front and back, the point is in the middle”), which continues 斲必半， “cutting must be by halves” .

One-ended schedule. Day k halves the remaining piece, of length $\tau 2^{-(k-1)}$, and spends $\frac{3}{4}C\tau^3 8^{-(k-1)}$. The total is $\frac{3}{4} \cdot \frac{8}{7} K_\tau = \frac{6}{7} K_\tau$; the pieces taken away keep $\sum_{k \geq 1} 8^{-k} K_\tau = \frac{1}{7} K_\tau$.

Alternating schedule (the reading favoured by 斲必半). One halving a day, taking the half alternately from the front and from the back. Each day is one halving of the remaining piece, so the spending is that of the one-ended schedule, $\frac{6}{7} K_\tau$; the remainder converges to the point $\frac{\tau}{2} + \frac{\tau}{8} + \dots = \frac{2\tau}{3}$.

Two quarter-cuts a day (a construction motivated by 前後取; the text does not fix the fractions). Day k cuts the remaining piece $r = \tau 2^{-(k-1)}$ at $s = \frac{1}{4}$ and $s = \frac{3}{4}$, keeping the middle half. The two cuts spend $C r^3 (1 - 2 \cdot \frac{1}{64} - \frac{1}{8}) = \frac{27}{32} C r^3$ (equivalently $\frac{9}{16} + \frac{9}{32}$ by two applications of Theorem 1), and the total is $\frac{27}{32} \cdot \frac{8}{7} K_\tau = \frac{27}{28} K_\tau$, with $\frac{1}{28} K_\tau$ kept in the pieces taken away. The contrast $\frac{6}{7}$ against $\frac{27}{28}$ is between one half-cut and two quarter-cuts a day, whichever ends are cut.

All three schedules converge to a point, the Mohist 端: the end, the point $2\tau/3$, and the midpoint. If 中 in 端中 means the midpoint, the explanation fits the symmetric schedule, while 斲必半 fits the alternating one. What each author is committed to, and what the theorem does with it:

- The dialecticians hold that the halving never ends. Theorem 2 agrees for the figure and adds a price: their schedule spends only $\frac{6}{7}$ of the action, since it never cuts again what it took away.
- The Mohists hold that the halving ends at a point. Theorem 2 agrees that each schedule converges to a point, and the point carries no action. The unhalvable 端 as a least magnitude is a reading the theorem leaves aside, as the Planck paper already records.
- Newton' s scholium closing Book I, Section I, takes the geometers' side (divisibility without end). Theorem 2 is that side made quantitative: refinement exhausts the action only when every piece is cut again.

The parallel with Zeno' s dichotomy (Aristotle, Physics VI.9) is recorded as convergence; transmission would need evidence this note does not have.

4. The weight of a cut

Take the Euclidean path measure of a free particle with variance rate $\hbar/m$: unconditioned centred Brownian motion X with $E[X_t^2] = \hbar t/m$, then pin it at the cell' s endpoints. Given the endpoints of a piece of length L , the value at fraction s is Gaussian with variance $\hbar s(1-s)L/m$, independent of the finer hats and of the other pieces, by the Markov property of the bridge (the Lévy construction, for any nested sequence). A

force F shifts the mean of that value by the sagitta $h = Fs(1-s)L^2/(2m)$. The Kullback–Leibler divergence of the shifted from the free law (the expected log-likelihood ratio, which here also equals the ratio evaluated at the shift) is

$$\frac{h^2}{2\hbar s(1-s)L/m} = \frac{mh^2}{2s(1-s)L\hbar} = \frac{3s(1-s)CL^3}{\hbar},$$

the spent share of Theorem 1 divided by $\hbar$. By the chain rule the divergences add over cuts, and along a sequence with mesh tending to zero they sum to $K_\tau/\hbar$, the Cameron–Martin exponent for the shift δx . The Cameron–Martin density factorizes over cuts, each contributing the normalizing factor $e^{-3s(1-s)CL^3/\hbar}$. This is the Euclidean counterpart of the real-time two-path phase K_τ/ε of the two-path note (the same number with ε in the role of $\hbar$), and of the product-over-cuts structure that § 7 of the fifth-postulate note reads in the time-step partition of Rivero 1998. For $\hbar \rightarrow 0$ every cut, however small, is decisive; for $\hbar > 0$ the cuts whose share falls below $\hbar$ carry little evidence (under one nat). $\square$

5. A floor counts the cuts

If every cut that a record exhibits must spend at least $\kappa > 0$, Theorem 2 gives at most K_τ/κ such cuts, for every schedule. For halving refinement the 2^n cuts of level n each spend $\frac{3}{4}K_\tau 8^{-n}$, so the exhibited levels are those with $8^n \leq 3K_\tau/(4\kappa)$.

Two floors must be kept apart: κ here floors the share a cut spends, $3s(1-s)K_L$, while the thresholds in the other notes floor the action of a cell, K_τ ; for halving the two differ by the factor $\frac{3}{4}$. The Planck paper’s eq. (2) is $K_\tau \geq 2z_{1-\epsilon}^2\hbar$: τ_* is the cell with $K_{\tau_*} = 2z_{1-\epsilon}^2\hbar$, whose halving spends $\frac{3}{2}z_{1-\epsilon}^2\hbar$. The round-3 resolution threshold $\tau \geq (24m\varepsilon d_{n,\eta}/F^2)^{1/3}$ is $K_\tau \geq \varepsilon d_{n,\eta}$, that is $\kappa = \frac{3}{4}\varepsilon d_{n,\eta}$ in cut-share form. Corollary 5 thus places a floor, when one is given, at a definite depth of the cut process; it supplies no floor.

5b. The mark mesh from the path measure (Proposition 6)

Keep the measure of § 4: unconditioned centred Brownian motion with $E[X_t^2] = \hbar t/m$, the Euclidean free-particle measure, whose diffusion coefficient $\hbar/2m$ is the one of Nelson (1966) (metadata). Assume $m, \tau, \hbar > 0$, $F \in \mathbb{R}$, equal priors and $0 < \epsilon < 1/2$. Let ℓ be a fixed endpoint chord and let B be the centred bridge with covariance $E[B_u B_v] = (\hbar/m)(\min(u, v) - uv/\tau)$. Compare $P_0 = \mathcal{L}(\ell + B)$ and $P_1 = \mathcal{L}(\ell - \delta x + B)$, where $\delta x(u) = Fu(\tau - u)/(2m)$. This specifies additive position noise around the two Newtonian comparison paths. Their Cameron–Martin norm is $d^2 = \|\delta x\|_{\text{CM}}^2 = (m/\hbar) \int \dot{\delta x}^2 = 2K_\tau/\hbar$. Writing $h = -\delta x$, the log-likelihood ratio is $Y - d^2/2$, where $Y = \langle h, X - \ell \rangle_{\text{CM}}$ denotes the Gaussian stochastic pairing, defined even though typical paths lie outside the Cameron–Martin space. Here integration by parts makes it the ordinary area statistic $Y = -(F/\hbar) \int_0^\tau (X_u - \ell_u) du$. Under P_0 , $Y \sim N(0, d^2)$; under P_1 , $Y \sim N(d^2, d^2)$. Thus the Neyman–Pearson test with equal priors thresholds that statistic halfway and errs with probability

$$p_{\text{err}} = \Phi\left(-\frac{1}{2}\|\delta x\|\right) = \Phi\left(-\sqrt{K_\tau/2\hbar}\right).$$

Hence $p_{\text{err}} \leq \epsilon$ iff $K_\tau \geq 2z_{1-\epsilon}^2\hbar$, that is $\tau\Delta E = 12K_\tau \geq 24z_{1-\epsilon}^2\hbar$ and, for $F \neq 0$, $\tau \geq \tau_* = (48z_{1-\epsilon}^2 m\hbar/F^2)^{1/3}$: eq. (2) of the Planck paper, including its “about $65\hbar$ at $\epsilon = 0.05$ ” ($2z_{0.95}^2 = 5.41$). A single cut observed alone gives $\Phi(-\sqrt{\text{share}/2\hbar})$ in the same way. $\square$

The Planck paper reaches (2) from real-time Gaussian marks whose record and recoil obey Robertson's bound $\hbar/2$; here the same constant comes from the Euclidean path measure $e^{-S/\hbar}$ through the Cameron–Martin theorem alone. The agreement is exact and structural. In Theorem 2 of the Planck paper, the mark trade-off $\kappa_{\text{mark}} = \hbar/2$ gives $\sup d_{\text{mark}}^2 = F^2\tau^3/(24m\kappa_{\text{mark}}) = 2K_\tau/\hbar$. Its optimization uses the same Dirichlet inverse of $-d^2/du^2$ with zero boundary values, hence the same parabolic extremizer and $\tau^3/12$. The full bridge attains this Gaussian information; the independent, uncorrelated, non-adaptive mark protocols approach it in the dense limit for tests invariant under initial position and velocity. This equality of sharp information retains the different experimental premises; it supplies no equivalence of all records or of Nelson's state-dependent dynamics. Both routes supply $\hbar > 0$, giving a second conditional derivation of the mesh, with the necessity question of the fifth-postulate note unchanged. What it adds is the reading of the floor on recording the inertial–parabola difference as the distinguishability threshold of the force in the free-particle path measure. At small ϵ , $z_{1-\epsilon}^2 \simeq 2\ln(1/\epsilon)$, so this floor grows like $4\hbar \ln(1/\epsilon)$ in cell action, the same logarithmic law as Theorem E of the fifth-postulate note, $ab \geq \hbar c_*(\epsilon)$ with $c_*(\epsilon) \simeq \frac{1}{2} \ln(1/\epsilon)$ in window half-widths; the constants refer to different quantities (a cell's Galileo action and a phase-space window).

5c. Mark protocols spend exactly the cut measure (Proposition 7)

Proposition 7 (2026-09-29, Claude, generalizing a single-cut computation by Fable; refereed by Fable, ACCEPT with wording fixes applied). Take the mark model of the Planck paper, § 3: mark j returns the position with Gaussian error δ_j and delivers a Gaussian impulse Δ_j with $\delta_j\Delta_j \geq \kappa$; protocols are non-adaptive and tests invariant under the unknown initial position and velocity, and marks with $\delta\Delta = \kappa$ are available at every $\delta > 0$ (as in the sharpness of Theorem 2; the upper bound needs only $\delta\Delta \geq \kappa$). Let the marks sit at $t_0 < t_1 < \dots < t_N$, with $\tau = t_N - t_0$ and pieces $\tau_i = t_i - t_{i-1}$. Then the supremum over the marks' δ_j, Δ_j of the discernibility is

$$\sup d^2 = \frac{K_\tau - \sum_i K_{\tau_i}}{\kappa} = \frac{1}{\kappa} \sum_{\text{cuts}} \frac{m}{2} \int \dot{\phi}_{\text{cut}}^2, \quad K_\tau = \frac{F^2\tau^3}{24m},$$

the action that the interior cuts spend by Theorems 1–2. It is approached, never attained, with sharp end marks and balanced interior marks. Special cases: a single cut at fraction s gives $3s(1-s)K_\tau/\kappa$; N equal pieces give $(1-N^{-2})K_\tau/\kappa$, so n halvings give $(1-4^{-n})K_\tau/\kappa$; the dense limit gives K_τ/κ , the paper's Theorem 2. Adding a mark inside a piece adds exactly that cut's share, $3s(1-s)K_{\tau_i}/\kappa$: the squared statistical distance is additive under refinement at every finite stage.

Proof. Write $T(w) = \sum_{t_i > w} u_i(t_i - w)$ for the test weights u . The invariance constraints $\sum u_i = \sum u_i t_i = 0$ say exactly that T is piecewise linear with kinks at the t_j and vanishes outside (t_0, t_N) , and every such T arises from one u , with u_j the jump of T' at t_j . The paper's § 3 gives the signal $\frac{F}{m} \int T$ and the noise $\sum_j \delta_j^2 u_j^2 + \sum_j \Delta_j^2 T(t_j)^2/m^2 \geq \frac{2\kappa}{m} \sum_j |u_j| |T(t_j)|$, by the arithmetic–geometric mean. Summation by parts gives $\sum_j u_j T(t_j) = -\int T'^2$, hence

$$d^2 \leq \frac{F^2}{2m\kappa} \frac{(\int T)^2}{\int T'^2}.$$

Let V_h be the piecewise-linear functions on the grid vanishing at t_0, t_N , and $T_h \in V_h$ the Galerkin solution of $-T'' = 1$: $\int T_h' \varphi' = \int \varphi$ for all $\varphi \in V_h$. Cauchy–Schwarz in the energy product gives $(\int T)^2 \leq \int T_h'^2 \int T'^2$ with $\int T_h'^2 = \int T_h$, so the ratio is at most $\int T_h$. In one dimension T_h interpolates $p(w) = \frac{1}{2}(w - t_0)(t_N - w)$ at the nodes, since the Dirichlet

Green' s function with its source at a node lies in V_h and the error $p - T_h$ is Galerkin-orthogonal to V_h . So $\int T_h$ is the trapezoid rule for p (nodal exactness of one-dimensional linear elements: Strang and Fix, *An Analysis of the Finite Element Method*, 1973; recalled), whose error on a piece is $-\tau_i^3 p''/12 = \tau_i^3/12$, and $\int T_h = (\tau^3 - \sum_i \tau_i^3)/12$. This gives the upper bound $F^2(\tau^3 - \sum \tau_i^3)/(24m\kappa)$. For the supremum take $T = T_h$, which is nonnegative and concave, so all $u_j T(t_j)$ have one sign; choose $\delta_j = \Delta_j T_h(t_j)/(m|u_j|)$ with $\delta_j \Delta_j = \kappa$ at interior marks, and let the end marks be sharp, $\delta_0, \delta_N \rightarrow 0$ with $\Delta_0, \Delta_N \rightarrow \infty$. Their impulses drop out of every invariant statistic, since $S_0 = S_N = 0$: the one at t_N reaches no later reading, and the one at t_0 acts as a shift of the unknown v_0 . The value is never attained, because $u_0, u_N \neq 0$ leave $\delta_0^2 u_0^2 + \delta_N^2 u_N^2 > 0$ at any finite end mark. Equivalently, $(F/m)T_h$ is the energy projection of the Galilean displacement onto the polygons on the grid, so $\sup d^2 = \frac{1}{\kappa} \frac{m}{2} \int (\Pi_h \delta x)^2$, which is Theorems 1–2 by construction. $\square$

The two constructions of this note and the paper' s are therefore one: the Newton cell action spent by any finite set of cuts is, over κ , the best discernibility of a mark protocol placed at those cuts. With $\kappa = \hbar/2$ the dense value reproduces Proposition 6' s exponent, $\frac{1}{2}\sqrt{K_\tau/\kappa} = \sqrt{K_\tau/(2\hbar)}$, from the mark model instead of the path measure. The continuity note reads the additivity as Leibniz' s petites perceptions.

The schedules of § 3 acquire a records meaning. Marks placed at the cuts of Hui Shi' s stick, however many, reach at most $\frac{6}{7}K_\tau/\kappa$, and the two quarter-cuts a day reach $\frac{27}{28}K_\tau/\kappa$: the stick that is “not exhausted in ten thousand generations” leaves one seventh of the discernibility of the comparison permanently out of reach, because the pieces it takes away are never marked again. The deficit exists only on the floor branch; with $\kappa = 0$ any three marks decide.

6. Consequence for STATE

Atlas § 1b: the Newton row of “cuts at any position” is proved here (Theorems 1–2, Corollary 3), with the Lévy–Ciesielski identification and the Euclidean weight of a cut (Proposition 4). Proposition 6 is now refereed: its Gaussian information equals the sharp mark information under the respective stated hypotheses. The necessity question is unchanged. On the gauge side of § 1b the free-field defect of a cut at fraction s is (5) scaled by $4s(1 - s)$ in its size bound (Corollary 2_s); the $U(1)$ Theorem 5 holds for every cut fraction (Corollary 5_s there).